

Teacher Guide: Mathematics

Period 1

Entry Ticket

Look at these two math statements: (1) $3x + 5$ (2) $3x + 5 = 11$. What is different about them? Write one sentence.

Teacher Script

Today we begin linear equations. An equation is a statement that two expressions are equal, joined by an '=' sign - unlike an expression, which has no equals sign and cannot be 'solved.' A linear equation in one variable has only one letter (variable) raised to the first power, like $3x + 5 = 11$. Here, x is the variable, 3 is its coefficient, 5 and 11 are constants. To solve, we use properties of equality: whatever we do to one side, we must do to the other - add, subtract, multiply, or divide equally - just like keeping a balance scale level. If we add weight to one pan, we must add the same weight to the other pan or it tips. This balance idea is the foundation for solving every equation we will meet this unit.

Blackboard Notes

- Expression: no '=' sign, e.g., $3x + 5$
- Equation: has '=' sign, e.g., $3x + 5 = 11$
- Linear equation in one variable: one variable, power 1
- Variable: unknown letter (x)
- Coefficient: number multiplying variable (3 in $3x$)
- Constant: fixed number (5, 11)
- Properties of Equality: add/subtract/multiply/divide same value both sides
- Balance analogy: equation = balanced scale

Mentor Moment

Think of an equation like a friendship - fair and balanced. If you take something away from one side without doing the same on the other, the relationship (equation) breaks. Keeping both sides equal, in math and in life, keeps things fair and true.

Homework

1. Convert the expression $5x - 3$ into an equation by making it equal to 12, then identify variable, coefficient, and constant.
2. Explain the subtraction property of equality using the balance analogy in your own words.

Period 2

Entry Ticket

If $x + 5 = 12$, what value of x makes this true? How did you figure it out?

Teacher Script

Yesterday we learned what a linear equation looks like. Today we solve them! Think of an equation like a balanced scale - whatever you do to one side, you must do to the other to keep it balanced. This is called the Property of Equality: you can add, subtract, multiply, or divide both sides by the same number without changing the truth of the equation. We use this to isolate the variable - get it alone on one side. For example, in $x + 5 = 12$, subtract 5 from both sides to get $x = 7$. For two-step equations like $2x + 3 = 11$, first subtract 3, then divide by 2. Always check your answer by substituting it back into the original equation - if both sides match, your solution is verified. Let's practice the balance method together, step by step, with integers including negatives.

Blackboard Notes

- Properties of Equality: add/subtract/multiply/divide same number on both sides
- Balance Method: keep both sides equal like a weighing scale
- One-step equation: $x + a = b \rightarrow x = b - a$
- Two-step equation: $ax + b = c \rightarrow x = (c-b)/a$
- Solution of an Equation: the value that makes it true
- Verification: substitute answer back into original equation
- Example: $x - 4 = 9 \rightarrow x = 13 \rightarrow$ check: $13 - 4 = 9$?

Mentor Moment

Mathematicians don't just find answers - they check their work. Verifying your solution by substitution is like a scientist double-checking an experiment. This habit builds confidence and catches careless mistakes, a skill useful far beyond math class.

Homework

1. Solve and check: $x - 6 = -2$
2. Solve and check: $4x + 1 = 17$

Period 3

Entry Ticket

Quick recall: If $x + 5 = 12$, what value of x makes both sides equal? How did you find it?

Teacher Script

Yesterday we solved equations by adding or subtracting the same number on both sides. Today we learn a shortcut called Transposition. Instead of writing 'add 5 to both sides', we simply move the term to the other side and flip its sign. For example, in $x + 5 = 12$, we transpose +5 to become -5 on the right: $x = 12 - 5$, so $x = 7$. Similarly, if a term is multiplied, we transpose it as division (and vice versa). Transposition is not magic - it is just a faster way of applying the properties of equality we already know. After solving, always verify by substituting the value back into the original equation to check both sides are equal. Watch carefully for sign errors - this is the most common mistake students make.

Blackboard Notes

- Transposition = shortcut for properties of equality
- Moving a term across '=' changes its sign: + becomes -, - becomes +
- Multiplication becomes division and division becomes multiplication when transposed
- Steps: (1) Identify terms with variable and constants (2) Transpose constants to one side (3) Transpose variable terms to other side (4) Simplify (5) Verify solution
- Example: $3x + 4 = 19 \rightarrow 3x = 19 - 4 \rightarrow 3x = 15 \rightarrow x = 5$
- Always perform same operation on both sides if transposition is unclear
- Verification: substitute solution back into original equation

Mentor Moment

Many mathematicians use shortcuts once they deeply understand the reasoning behind them. Transposition looks like a trick, but it's built entirely on the equality principle you already know. Trust the process, but never skip verification - even experts check their work before trusting an answer.

Homework

1. Solve using transposition: $7x + 4 = 25$, and verify your solution.
2. Explain in your own words why transposition works, linking it to the properties of equality.

Period 4

Entry Ticket

Solve: $2x + 5 = 15$. Then look at this new equation: $3x + 5 = x + 15$. Can you solve it the same way? What is different?

Teacher Script

Today we solve equations that have the variable on BOTH sides, like $3x + 5 = x + 15$. Unlike before, we cannot just move numbers - we must first collect all variable terms on one side and all constants on the other.

This is called transposition: when a term crosses the equal sign, its operation reverses (addition becomes subtraction, and vice versa). Remember the golden rule from before - whatever we do to one side, we must do to the other, so transposition is just a shortcut for that. After transposing, solve like a simple equation. Finally, always check your solution by substituting the value back into BOTH the left-hand side and right-hand side separately - if $LHS = RHS$, your answer is correct. This checking step catches sign errors, a very common mistake. Watch carefully for incorrect sign changes and forgetting to apply an operation to both sides - these are the two biggest mistakes students make with this topic.

Blackboard Notes

- Equation with variables on both sides: e.g., $3x + 5 = x + 15$
- Transposition: moving a term across '=' flips its sign/operation
- Step 1: Collect variable terms on one side
- Step 2: Collect constant terms on the other side
- Step 3: Simplify and solve for the variable
- Step 4: Check - substitute value into LHS and RHS separately
- Solution is correct only if $LHS = RHS$
- Common errors: wrong sign change, operating on only one side

Mentor Moment

Think of transposition like moving furniture across a room divider - if a couch 'crosses over,' it doesn't stay the same shape; it flips orientation. Similarly, a term crossing the equal sign flips its operation. Always double-check your solved equation by plugging the value back in - professionals in engineering and finance verify their calculations the same way before trusting them.

Homework

1. Solve: $5x + 7 = 2x + 25$, and verify your answer by substitution.
2. A student wrote: ' $3x + 4 = x + 10$, so $3x + x = 10 + 4$ ' - explain the sign error in this transposition and solve the equation correctly.

Period 5

Entry Ticket

Write one equation for: 'A number increased by 5 equals 12.' What is the number?

Teacher Script

Today we apply everything we've learned about linear equations to real-life word problems. The key skill is translation: turning words into an equation using a variable for the unknown quantity. We'll identify key phrases like 'sum,' 'difference,' 'twice,' 'consecutive,' and 'age of' to build equations, sometimes with variables

on both sides. After solving using transposition and properties of equality, we must always check: does this answer make sense in the real-world context? For example, a person's age can't be negative, and the number of items can't be a fraction. We'll practice translating, solving, and verifying together, then you'll create your own word problem for a partner to solve, including at least one with variables on both sides.

Blackboard Notes

- Steps: Read -> Identify unknown -> Assign variable -> Write equation -> Solve -> Verify
- Key phrases: sum(+), difference(-), twice($2x$), consecutive(x , $x+1$, $x+2$)
- Example: 'Sum of two consecutive numbers is 25' -> $x + (x+1) = 25$
- Variables both sides: $5x + 2 = 3x + 10$
- Always check solution fits the real-world situation
- Solution of an equation = value that makes both sides equal

Mentor Moment

Word problems can feel intimidating, but they're just puzzles in disguise! The real skill isn't math-it's careful reading and translation. Once you can 'speak equation,' you'll notice these patterns everywhere: in shopping discounts, sports scores, even recipes. Trust the process: translate, solve, check.

Homework

1. Create your own word problem involving variables on both sides, then solve it and check the answer makes sense.
2. The perimeter of a rectangle is 40 cm; the length is 4 cm more than the width. Find the dimensions using an equation.